



UNIVERSITY OF THE FREE STATE

BLOEMFONTEIN CAMPUS

CSIS3744
MEMORANDUM

**DEPARTMENT OF COMPUTER SCIENCE AND
INFORMATICS**

CONTACT NUMBER: 051 401 3702

EXAMINATION: Special Examination 2021

ASSESSOR: 1. Prof Okuthe P. Kogeda

MODERATOR: 1. Mnr. W.S.J. Marais

TIME: 3 HOURS

MARKS: 100

INSTRUCTIONS: Answer ALL the questions. Write legibly. Number your questions clearly and correctly.

Question 1 [28 Marks]

Gamestore traders want to divide their network into two subnets. Their current network address is 192.168.0.0.

- (a) To which network class does the above mentioned network address belong? [1 Mark]

<<< Class C ✓ >>>

- (b) Calculate the subnet mask for Gamestore traders. [3 Marks]

<<< Default subnet mask for class C network is 255.255.255.0.

Binary value of subnet mask is:

11111111.11111111. 11111111.00000000 ✓

$$2^n = 2 \quad n = 1 \quad \checkmark$$

Change 1 bit from 0 to 1 for the last octet:

11111111.11111111.11111111.10000000

Value of 10000000 -> $2^7 = 128$

Thus new subnet mask is 255.255.255.128 ✓

OR

Default subnet mask for class C network is 255.255.255.0.

Binary value of subnet mask is 11111111.11111111. 11111111.00000000

$$2^n - 2 = 2 \quad n = 2 \quad \checkmark$$

Change 2 bits from 0 to 1 for the last octet.

11111111.11111111.11111111.11000000

Value of 11000000 -> $2^7 + 2^6 = 128 + 64 = 192$ ✓

The new subnet mask is 255.255.255.192 ✓

- (c) How many machines will the whole network be able to host? [3 Marks]

$$<<< 2^7 - 2 \checkmark = 128 - 2 = 126 \times 2 \checkmark = 252 \checkmark$$

OR

$$(2^6 - 2) \checkmark = 62 \times 2 \checkmark = 124 \checkmark$$

- (d) Complete the table below by supplying each subnet's network address, broadcast address and usable host address range: [4 Marks]

Subnet	Network Address	Broadcast Address	Usable Host Address Range
1	192.168.0.0 ¹ / ₂	192.168.0.127 ¹ / ₂	192.168.0.1 through ¹ / ₂ 192.168.0.126 ¹ / ₂
2	192.168.0.128 ¹ / ₂	192.168.0.255 ¹ / ₂	192.168.0.129 through ¹ / ₂ 192.168.0.254 ¹ / ₂

OR

	Subnet	Network address	Broadcast address	Usable host address range
(i)	1	192.168.0.64 ¹ / ₂	192.168.0.127 ¹ / ₂	192.168.0.65 ¹ / ₂ to 192.168.0.126 ¹ / ₂
(ii)	2	192.168.0.128 ¹ / ₂	192.168.0.191 ¹ / ₂	192.168.0.129 ¹ / ₂ to 192.168.0.190 ¹ / ₂

- (e) Explain the disadvantages that the implementation of multiple virtual servers on a single host machine might hold. [4 Marks]

<<< *Compromised performance: When multiple VMs contend for finite physical resources, one VM could monopolize those resources and impair the performance of other VMs on the same computer. ✓

*Increased complexity: Although virtualization reduces the number of physical machines to manage, it increases complexity and administrative burden in other ways. ✓

*Increased licensing costs: Because every instance of commercial software requires its own license, every VM that uses such software comes with added cost. ✓

*Single point of failure: If a host machine fails, all its guest machines will fail too. ✓

>>>

- (f) What are some of the different reasons to use VLANs? [4 Marks]

<<< Reasons for using VLANs include:

- * You wish to isolate connections with heavy or unpredictable traffic patterns, such as when separating heavy VoIP traffic from other network activities. ✓
- * You want to identify groups of devices whose data should be given priority handling, such as executive client devices or an ICS (industrial control system) that manages a refrigeration system or a gas pipeline. ✓

- * You want to contain groups of devices that rely on legacy protocols incompatible with the majority of the network's traffic, such as a legacy SCADA (supervisory control and data acquisition) system monitoring an oil refinery. ✓
- * You wish to separate groups of users who need special or limited security or network functions, such as when setting up a guest network. ✓
- * You are configuring temporary networks, such as when making specific network resources available to a short-term project team. ✓
- * You intend to reduce the cost of networking equipment, such as when upgrading a network design to include additional departments or new types of network traffic. ✓>>>

(g) What is the Point-to-Point Protocol (PPP) and explain how it works? [2 Marks]

<<< PPP (Point-to-Point Protocol) is a Data Link layer protocol that directly connects two WAN endpoints. ✓ One example might be when a DSL or cable modem connects to a server at the ISP. PPP headers and trailers create a PPP frame that encapsulates Network layer packets. The frames total only 8 or 10 bytes, the difference depending on the size of the FCS field (recall that the FCS field ensures the data is received intact). ✓>>>

(h) Name and explain two approaches likely to be used by a hacker when attempting to perform a VLAN hopping attack. [3 Marks]

<<< The two approaches the hacker might use for VLAN hopping are as follows:

- * double tagging ¹/₂ - The hacker stacks VLAN tags in Ethernet frames. When the first, legitimate tag is removed by a switch, the second, illegitimate tag is revealed, tricking a switch into forwarding the transmission on to a restricted VLAN. ✓
- * switch spoofing ¹/₂ - An attacker connects to a switch and then makes the connection look to the switch as if it's a trunk line. The switch might autoconfigure its port into trunk mode when it detects trunk mode on the other end of the connection. A hacker can then feed his own VLAN traffic into that port and access VLANs throughout the network. ✓>>>

(i) Explain the difference between how a switch and a hub operates. [2 Marks]

<<< A hub receives a signal via one port and sends (broadcasts) the received signal to all the other ports. ✓ A switch receives a signal via one port and sends

the received signal only to the destination port according to the specified MAC address contained in the message. ✓>>>

- (j) Explain the difference between a default VLAN and a native VLAN. [2 Marks]

<<< A default VLAN is typically preconfigured on a switch and initially includes all the switch's ports. Other VLANs might be preconfigured as well, depending on the device and manufacturer. The default VLAN cannot be renamed or deleted; however, ports in the default VLAN can be reassigned to other VLANs. ✓ A native VLAN receives all untagged frames from untagged ports. By default, this is the same as the default VLAN. However, this configuration poses a security risk when untagged traffic is allowed to travel in a VLAN-managed network. To protect the network from unauthorized traffic, the native VLAN should be changed to an unused VLAN so that untagged traffic essentially runs into a dead-end. Each switch port can be configured for a different native VLAN. However, switch ports on each end of a trunk should agree on the native VLAN assignment. If the ports don't agree, this is called a native VLAN mismatch, or just VLAN mismatch, and will result in a configuration error. ✓>>>

Question 2 [30 Marks]

TCP and UDP are the major protocols used on the transport layer of the OSI model.

- (a) What is the responsibility of the presentation layer of the OSI model? [1 Mark]

<<< Responsible for reformatting, compressing, and/or encrypting data in a way that the receiving application can read. ✓>>>

- (b) A Windows computer on your network is having issues resolving queries for a specific domain, but not other domains. Other computers on the same network resolve the name just fine. What command can you issue that might fix the problem? [1 Mark]

<<< ipconfig/flushdns ✓>>>

- (c) You realized that the network's performance has degraded as a result of the use of TCP on the transport layer. You consult and you are advised to change to UDP. Explain how this change would improve the performance of the network. [3 Marks]

<<< Candidates are expected to state that: UDP adds less payload (8 bytes while TCP adds 20 bytes); ✓ UDP is connectionless oriented; ✓ UDP has no checksum for errors; ✓ etc. >>>

- (d) Discuss the main features of proxy server that makes it provide security features to a network. [3 Marks]

<<< Any three of: Proxy server acts as an intermediary between external and internal networks; ✓ Screens all incoming and outgoing traffic; ✓ Manages security at Application layer; ✓ Appears as an internal network server to the outside world, but is a filtering device for internal LAN; ✓ One of its most important functions is preventing the outside world from discovering the addresses of the internal network. ✓ >>>

- (e) Discuss the difference between IPv4 and IPv6. [3 Marks]

<<< *IPv4 is a 32-Bit IP Address. IPv6 is 128 Bit IP Address. ✓

*IPv4 is a numeric address, and its binary bits are separated by a dot (.) IPv6 is an alphanumeric address whose binary bits are separated by a colon (:). ✓

*Without NAT, IPv6 is faster than IPv4 ✓ >>>

- (f) Which protocols on IPv4 networks perform the ICMPv6 functions on IPv6 networks? [1 Mark]

<<< ICMP and ARP ✓ >>>

- (g) Which protocol can provide VoIP services over a Metro Ethernet connection?

[1 Mark]

<<< SIP ✓ >>>

- (h) IPv6 has *global* and *link local* unicast addresses. Explain the difference between these two types of unicast addresses. [4 Marks]

<<< IPv6 unicast addresses come in two types:

* global addresses-Can be routed on the Internet and is similar to public IPv4 addresses. Most begin with the prefix 2000::/3, although other prefixes are being released. The /3 indicates that the first three bits are fixed and are always 001. ✓✓

* link local addresses-Can be used for communicating with nodes in the same link, and is similar to an autoconfigured APIPA address in IPv4. It begins with FE80::/10. The first 10 bits of the reserved prefix are fixed (1111 1110 10), and the remaining 54 bits in the 64-bit prefix are all zeroes. Therefore, a link local address prefix is sometimes written as FE80::/64. Link local addresses are not allowed past the local link or on the Internet. ✓✓ >>>

- (i) An organisation may configure a DNS server as *primary*, *secondary*, *caching* and *forwarding* DNS server for use on their network. Explain these four different DNS servers. [4 Marks]

<<< An organization might utilize these four common types of DNS server configurations:

- * primary DNS server-The authoritative name server for the organization, which holds the authoritative DNS database for the organization's zones. This server is contacted by clients, both local and over the Internet, to resolve DNS queries for the organization's domains. ✓

- * secondary DNS server-The backup authoritative name server for the organization. When a secondary DNS server needs to update its database, it makes the request to the primary server for the update; this process is called a zone transfer. ✓

- * caching DNS server-A server that accesses public DNS data and caches the DNS information it collects. This server receives DNS queries from local network clients and works to resolve them by contacting other DNS servers for information. Caching DNS servers do not store zone files (which is why they must rely on their caches and resolution efforts), and therefore do not participate in zone transfers. ✓

- * forwarding DNS server-An optional server that receives queries from local clients but doesn't work to resolve the queries. Typically, a forwarding server will maintain its own DNS cache from previous queries, and so it might already have the information the client needs. If not, the forwarding server forwards the query to another server to resolve. Several forwarding servers might be strategically placed throughout the organization's network to reduce network traffic on slow links. ✓

>>>

- (j) What are the different categories of routers, and how do they compare? [3 Marks]

<<< Some of the different categories of routers are as follows:

- * Core routers, also called interior routers, are located inside networks within the same autonomous system. An AS (autonomous system) is a group of networks, often on the same domain, that are operated by the same organization. An AS is sometimes referred to as a trusted network because the entire domain is under the organization's control. Core routers communicate only with routers within the same AS. ✓

- * Edge routers, or border routers, connect an autonomous system with an outside network, also called an untrusted network. For example, the router that connects a business with its ISP is an edge router. ✓

- * Exterior router refers to any router outside the organization's AS, such as a router on the Internet backbone. Sometimes a technician might refer to her own edge router as an exterior router because it communicates with routers outside the AS. But keep in mind that every router communicating over the Internet is an edge

router for some organization's AS, even if that organization is a large telecommunications company managing a portion of the Internet backbone. ✓>>>

- (k) There are several interior gateway protocols, but only one current exterior gateway protocol. What is this protocol, and what characteristics does it have? [4 Marks]

<<< The Border Gateway Protocol (BGP) is the only current exterior gateway protocol, and has been dubbed the "protocol of the Internet." Whereas OSPF and IS-IS scouting parties only scout out their home territory, a BGP scouting party can go cross-country. BGP spans multiple autonomous systems and is used by edge and exterior routers on the Internet. ✓ Here are some special characteristics of BGP:

- * path-vector routing protocol-Communicates via BGP-specific messages that travel between routers over TCP sessions. ✓
- * efficient-Determines best paths based on many different factors. ✓
- * customizable-Can be configured to follow policies that might, for example, avoid a certain router, or instruct a group of routers to prefer one particular route over other available routes. ✓>>>

- (i) How is the TTL (Time to Live) field utilized in IPv4? [2 Marks]

<<< The TTL field indicates the maximum duration that the packet can remain on the network before it is discarded. Although this field was originally meant to represent units of time, on modern networks it represents the number of times a packet can still be forwarded by a router, or the maximum number of router hops it has remaining. ✓The TTL for packets varies and can be configured; it is usually set at 32 or 64. Each time a packet passes through a router, its TTL is reduced by 1. When a router receives a packet with a TTL equal to 0, it discards that packet and sends a TTL expired message via ICMP back to the source host. ✓>>>

Question 3 [30 Marks]

Structured cabling describes the best ways to install networking media to maximize performance and minimize upkeep.

- (a) In considering a telecommunications service provider's network, what is the demarc, and where is it located? [2 Marks]

<<< In general, the device that marks where a telecommunications service provider's network ends and the organization's network begins is the demarc, or demarcation point. For example, an ISP (Internet service provider) might be responsible for fiber-optic cabling to your building to connect to your LAN. The device where the WAN ends and the LAN begins is the demarc. ✓The service provider is responsible for its network beyond the demarc, and, in most cases, the

organization is responsible for devices and services on the campus side of the demarc. ✓>>>

(b) What steps should be taken to manage documentation at a cabling plant?[4 Marks]

<<< Follow these guidelines to manage documentation at your cabling plant:

*Keep your cable plant documentation in a centrally accessible location. Make sure it includes locations, installation dates, lengths, and grades of installed cable. ✓

*Label every data jack or port, patch panel or punch-down block, and connector or circuit. ✓

*Use color-coded cables for different purposes and record the color schemes in your documentation. Cables can be purchased in a variety of sheath colors. For example, you might want to use pink for patch cables, green for horizontal wiring, purple for DMZ lines, and gray for vertical (backbone) wiring. ✓

*Be certain to update your documentation as you make changes to the network. The more you document, the easier it will be to move or add cable segments in the future. ✓>>>

(c) Explain the difference between single mode fiber and multimode fiber. [4 Marks]

<<< *SMF (single mode fiber) consists of a narrow core of 8 to 10 microns in diameter. Laser-generated light travels a single path over the core, reflecting very little. Because it reflects little, the light does not disperse as the signal travels along the fiber. This continuity allows SMF to accommodate the highest bandwidths and longest distances (without requiring repeaters) of all network transmission media. ✓✓

*MMF (multimode fiber) contains a core with a larger diameter than SMF, usually 50 or 62.5 microns, over which many pulses of light generated by a laser or LED light source travel at various angles. Signals traveling over multimode fiber experience greater attenuation than those traversing single mode fiber. Therefore, MMF is not suited to distances longer than a few kilometers. On the other hand, MMF is less expensive to install and, therefore, typically used to connect routers, switches, and servers on the backbone of a network or to connect a desktop workstation to the network. ✓✓>>>

(d) Explain the THREE different types of wireless network topologies that can be created. [3 Marks]

<<< Some of the different types of wireless topologies are as follows:

* ad hoc - A small number of nodes closely positioned transmit directly to each other without an intervening connectivity device. ✓

* infrastructure - An intervening connectivity device, a WAP (wireless access point) or AP (access point), accepts wireless signals from multiple nodes and retransmits them to the rest of the network. To cover its intended range, an access point must have sufficient power and be strategically placed so that all connected nodes can communicate with it. ✓

* mesh - Several access points work as peer devices on the same network, where the AP devices cooperate to provide more fault tolerant network access to clients. A wireless controller might be used only initially to configure the APs, or the APs might remain connected to the wireless controller for continued management. ✓

>>>

- (e) The configuration of wireless clients for wireless access to the network in an enterprise environment involve *on-boarding* and *off-boarding* processes. Explain these two-part process. [4 Marks]

<<< * on-boarding-Users or network technicians install a specific app, called an agent, on a user's device, whether the device is a smartphone, laptop, or tablet. This gives the device trusted access to certain portions of the network. Access to email services, file-sharing services, and certain network administrative features might all be controlled by the device's permission levels enabled by on-boarding that device. ✓✓

* off-boarding-The reverse procedure involves removing the agent. For security purposes, network administrators need a feature that allows them to do this remotely, in case a device is lost or stolen. This feature, called a remote wipe, clears a device of all important information, permissions, and apps without having physical access to the device. It might even allow you to completely disable the device, making any network or data access impossible. ✓✓>>>

- (f) Hackers are categorized according to their intent and the prior approval of the organizations whose networks they're hacking. Explain the difference between *white hat hacker* and *black hat hacker*. [2 Marks]

<<< *White hat hacker-These IT security experts are hired by organizations to assess their security and risks. They're sometimes called ethical hackers. Their goal is to identify security vulnerabilities of all kinds so the organization can make

changes to increase their security. The extent of their efforts is usually clearly defined in a written contract before they begin their testing, and their activities are limited by existing laws and restrictions. At no point is private data compromised outside of that trusted relationship. ✓

* black hat hacker-These groups or individuals use their skills to bypass security systems to cause damage, steal data, or compromise privacy. They're not concerned with legal restrictions, and are intent on achieving personal gain or executing a personal agenda against an individual or an organization. Some black hat hackers and groups are also available for hire to serve someone else's agenda.

✓>>>

(g) Explain the difference between hashing and encryption. [2 Marks]

<<< Hashing means to transform data through an algorithm that generally reduces the amount of space needed for the data. Hashing is not the same thing as encryption, though it's often listed as a type of encryption and does, in a similar manner, transform data from one format to another. ✓ Encrypted data can be decrypted, but hashed data cannot. Hashing is mostly used to ensure data integrity-that is, to verify the data has not been altered, which is similar to the purpose of a checksum. ***Encryption is a process that encodes a message or file so that it can only be read by certain people. Encryption uses an algorithm to scramble, or encrypt, data and then uses a key for the receiving party to unscramble, or decrypt, the information.*** ✓>>>

(h) In a Network Access Control (NAC) system, we have *nonpersistent* and *persistent* types of software agents that are commonly used. Explain these two. [2 Marks]

<<<*A nonpersistent agent, or dissolvable agent, remains on the device long enough to verify compliance and complete authentication, and then uninstalls. Devices might be required to periodically reinstall the agent to complete the authentication process again. ✓

* A persistent agent is permanently installed on a device. This more robust program might provide additional security measures, such as remote wipe, virus scans, and mass messaging. ✓>>>

- (i) Explain any FOUR different types of power flaws that can cause damage to your equipment. [4 Marks]

<<< The following list describes power flaws that can damage your equipment:

- * surge-A momentary increase in voltage due to lightning strikes, solar flares, or electrical problems. Surges might last only a few thousandths of a second, but can degrade a computer's power supply. Surges are common. You can guard against surges by making sure every computer device is plugged into a surge protector, which redirects excess voltage away from the device to a ground, thereby protecting the device from harm. Without surge protectors, systems would be subjected to multiple surges each year. ✓

- * noise-Fluctuation in voltage levels caused by other devices on the network or EMI. Some noise is unavoidable on an electrical circuit, but excessive noise can cause a power supply to malfunction, immediately corrupting program or data files and gradually damaging motherboards and other computer circuits. If you've ever turned on fluorescent lights or a microwave oven and noticed the lights dim, you have probably introduced noise into the electrical system. Power that is free from noise is called clean power. To make sure power is clean, a circuit must pass through an electrical filter. ✓

- * brownout-A momentary decrease in voltage; also known as a sag. An overtaxed electrical system can cause brownouts, which you might recognize in your home as a dimming of the lights. Such voltage decreases can cause computers or applications to fail and potentially corrupt data. ✓

- * blackout-A complete power loss. A blackout could cause significant damage to your network. For example, if a server loses power while files are open and processes are running, its NOS might be damaged so extensively that the server cannot restart and the NOS must be reinstalled from scratch. A backup power source, however, can provide power long enough for the server to shut down properly and avoid harm. ✓>>>

- (j) Describe the Synchronous Optical Network standard and provide its two key strengths. [3 Marks]

<<< SONET (Synchronous Optical Network) is a high-bandwidth WAN signalling technique developed for fiber-optic cabling by Bell Communications Research in the 1980s, and later standardized by ANSI and ITU. SONET specifies framing and multiplexing techniques at the Physical layer of the OSI model. ✓

Its four key strengths are that it:

- * Can integrate many other WAN technologies ✓

- * Offers fast data transfer rates ✓

- * Allows for simple link additions and removals ✓

- * Provides a high degree of fault tolerance. ✓>>>

Question 4 [12 Marks]

There are three primary areas or classifications of security controls. These include management security, operational security, and physical security controls.

- (a) Explain how a stateless firewall operates. [1 Mark]

<<< A stateless firewall manages each incoming packet as a stand-alone entity, without regard to currently active connections. ✓ >>>

- (b) Why do network administrators create domain groups to manage user security privileges? [1 Mark]

<<<To simplify the process of granting rights to users. ✓ >>>

- (c) What is the purpose of implementing a demilitarized zone [1 Mark]

<<< A demilitarized zone (DMZ) is a perimeter network that protects an organization's internal local-area network (LAN) from untrusted traffic. The end goal of a DMZ is to allow an organization to access untrusted networks, such as the internet, while ensuring its private network or LAN remains secure. ✓ >>>

- (d) Why is a BPDU filter needed at the demarc? [2 Marks]

<<< The ISP's STP-related topology information shouldn't be mixed with a corporate network's STP-related topology information. ✓✓>>>

- (e) What characteristic of ARP makes it particularly vulnerable to being used in a DoS attack? [1 Marks]

<<< ARP performs no authentication. ✓ >>>

- (f) Explain two security benefits of using virtualization on a network. [2 Marks]

<<< Fault and threat isolation; ✓ Simple backups, recovery, and replication. ✓ >>>

- (g) How does the CCMP help ensure data confidentiality? [2 Marks]

<<< CCMP helps ensure data confidentiality with both encryption and packet authentication by providing:

- * message integrity-CCMP uses CBC-MAC, which ensures incoming packets are, in fact, coming from their declared source, and does so using the block cipher algorithm AES, described next. ✓

- * encryption-CCMP also uses AES (Advanced Encryption Standard), which provides faster and more secure encryption than TKIP for wireless transmissions.

AES relies on a more sophisticated family of ciphers along with multiple stages of data transformation. ✓>>>

(h) Explain how the ATM is able to guarantee a specific QoS for certain types of transmissions? [2 Marks]

<<< Establishing a reliable connection allows ATM to guarantee a specific QoS for certain types of transmissions. ATM networks can supply four QoS levels, from a "best effort" attempt for noncritical data to a guaranteed, real-time transmission for time sensitive data. This is important for organizations using networks for time-sensitive applications, such as video and audio transmissions. ✓ For example, a company might want to use its ATM connection between two offices located at opposite sides of a state to carry voice phone calls with the highest possible QoS. On the other hand, the company might assign a low QoS to routine email messages exchanged between the two offices. ✓>>>

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